Supplementary materials for Reconfigurable Topological Photonic Crystal

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Topological properties of the structure

Let us consider a 2D photonic crystal (PC) with a structure of a honeycomb lattice of infinitely long silicon pillars with diameter d (refractive index of silicon is taken to be 3.5) surrounded by a uniform material with the refractive index n_{bg} schematically presented in Fig. 1(a) in the main text of the manuscript. The lattice constant of this honeycomb lattice is equal to a and the unit cell containing two pillars is schematically shown in Fig. S1(a) (dashed lines). In our 2D model, the PC structure is invariant along the z-axis, and we only analyze the behavior of the transverse-magnetic (TM) modes because, simple geometrical modifications of the crystal lattice allow for introduction of topological properties for this light polarization [1]. It is important to notice that the honeycomb lattice described above is equivalent to a larger triangular lattice containing six silicon pillars in a single unit cell. The dimensions of the unit cell of the triangular lattice are: the lattice constant $a_0 = \sqrt{3}a$, the spacing between the unit cell center and the center of the pillar $R = a_0/3$, and the pillar diameter used here is $d = 2/9a_0$. Description of the system in terms of a larger triangular lattice is required to characterize the topologically trivial and non-trivial PC geometries.

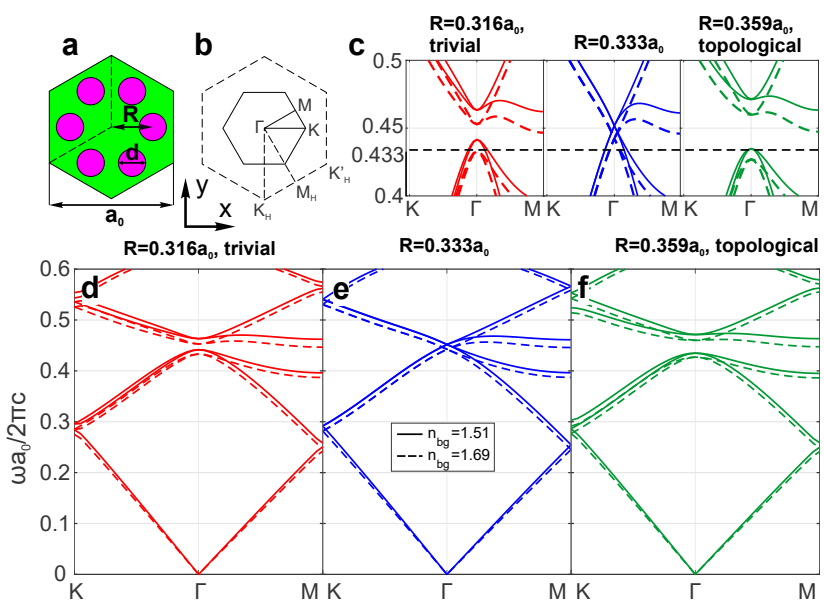

FIG. S1. (a) Schematic illustration of the unit cells of the triangular lattice (shaded in green) and the honeycomb lattice (dashed lines) with the geometrical parameters specified in the text. (b) The first Brillouin zones of the triangular lattice (solid lines) and the honeycomb lattice (dashed lines). The high symmetry points of the triangular lattice are $\Gamma = [k_x^{(\Gamma)}, k_y^{(\Gamma)}] = [0, 0]$, $K = [4\pi/(3a_0), 0]$, and $M = [\pi/a_0, \pi/(\sqrt{3}a_0)]$. High symmetry points of the honeycomb lattice are, $K_H = [0, -4\pi/(\sqrt{3}a_0)]$, $M_H = [\pi/a_0, -\sqrt{3}\pi/a_0]$, and $K'_H = [2\pi/a_0, -2\pi/(\sqrt{3}a_0)]$. (f)–(h) Full photonic band structures, and zooms on the band-gap regions (c)–(e) of the honeycomb lattice (triangular lattice with $R = a_0/3$—blue) exhibiting symmetry protected double Dirac cones (d), (g); triangular lattice with $R = 0.316a_0$ (red) and opened trivial gap (c), (f); and topological triangular lattice with $R = 0.359a_0$ (green) with non-trivial gap (e), (h). Solid lines and dashed lines correspond to the background refractive indices of 1.51 and 1.69, respectively.

Band diagrams for the TM modes of the structure discussed above were calculated using the plane-wave-expansion (PWE) method for background refractive indices of 1.51 and 1.69, and are presented in Figs. S1(d) and (g). The allowed eigen-frequencies ω of the structure are shown in function of the wavevectors lying along the high-symmetry directions in the first Brillouin zone (BZ) of the large triangular structure shown in Fig. S1(b). The Floquet periodicity used for PC analysis assumes the phase difference $\phi = \vec{k}a_n$ on the opposite unit cell boundaries, where a_n denotes one of the lattice vectors. For the periodicity along the x-direction, this phase can be expressed as $\phi = k_x a_0 = (k_x^{(1^{st}\mathrm{BZ})} + \frac{2\pi}{a_0}m)a_0$. Here, k_x denotes any of the allowed k-vectors, $k_x^{(1^{st}\mathrm{BZ})}$ is its equivalent from the first BZ, and $m \in [0, 1, 2, \dots]$ denotes the number of full field oscillations in the unit cell. We observe the presence of a doubly-degenerate Dirac cone at the Γ point, which is a result of folding of the bands from the larger BZ corresponding to the smaller unit cell of the honeycomb lattice [shown by dashed lines in Fig. S1(b)] [2]. The Dirac cones are located at points K_H and K'_H of the honeycomb lattice BZ. The doubly-degenerate Dirac cones are the consequence of the C_6 symmetry of the honeycomb lattice [1].

Now, we modify the geometry of the PC by varying the pillar spacing R. This transformation does not affect the C_6 system symmetry but can lead to changes in the topological properties of the PC band structure. For values of $R \neq a_0/3$, the lattice can no longer be described as a honeycomb lattice, and therefore, introduction of the triangular lattice with a larger unit cell was necessary. Variations of R lead to a spectral gap opening at the Γ point, where the Dirac cones were located. Although the band-gap is opened regardless of whether we increase or decrease the pillar spacing, the topological character of the gap is different in both cases. Smaller values of R result in weaker coupling between the adjacent meta-molecules and the topology of the system remains trivial. On the contrary, increased values of R result in stronger inter-cell coupling and the gap has a non-trivial topological character (see next section of the Supplementary materials). The band diagrams for the topologically trivial system with $R = 0.316a_0$, and for the non-trivial configuration with $R = 0.359a_0$ are shown in Figs. S1(c), (f) and S1(e), (h), respectively.

The conducting topological edge states exist at an interface between the materials with overlapping band-gaps but different topological orders, defined by the topological invariant known as Chern number [3]. The number of edge states, residing inside the bulk band-gap, is determined by the difference between Chern numbers of the two materials. The Chern number of the crystal lattice studied here can be evaluated using Hamiltonians derived from $k \cdot P$ theory [1, 4] or tight-binding approximation [2]. In the following section, we present the tight-binding approach to the Chern number calculation.

Tight-binding model

The photonic crystal considered in the main text of the manuscript is composed of a unit cell made of six silicon pillars, referred later as a meta-molecule (MM). For the TM-modes of the system, the z-component of the electric field ψ in the unit cell can be represented as a combination of Wannier states:

$$\psi = \sum_{X=A,B\dots F} A_X \psi_X e^{i\vec{k}\cdot\vec{r}}, \tag{S.0}$$

where ψ_X is the field distribution localized at the atom X, A_X is the corresponding complex amplitude, $\vec{k} = [k_x, k_y]$ is the light wavevector, and $\vec{r} = (x, y)$ denotes the position.

In the matrix notation, the electric field distribution in our system can be represented using a six-element column-vector. The elements of the vector represent the field amplitudes at the corresponding atoms:

$$\psi \to |\psi\rangle = \left[A_A, A_B, A_C, A_D, A_E, A_F\right]^T. \tag{S.1}$$

In this notation, the Wannier basis consists of the following six vectors:

$$|A\rangle = \begin{bmatrix}1\\0\\0\\0\\0\\0\end{bmatrix}, |B\rangle = \begin{bmatrix}0\\1\\0\\0\\0\\0\end{bmatrix}, |C\rangle = \begin{bmatrix}0\\0\\1\\0\\0\\0\end{bmatrix}, |D\rangle = \begin{bmatrix}0\\0\\0\\1\\0\\0\end{bmatrix}, |E\rangle = \begin{bmatrix}0\\0\\0\\0\\1\\0\end{bmatrix}, |F\rangle = \begin{bmatrix}0\\0\\0\\0\\0\\1\end{bmatrix}, \tag{S.2}$$

and the wavefunction is written as:

$$|\psi\rangle = \sum_{X=A,B\dots F} A_X |X\rangle. \tag{S.3}$$

The interaction Hamiltonian of the system can be described using nearest-neighbor hopping potentials. In this model, the interaction Hamiltonian H_W (written in the basis of Wannier states) can be separated into the intra-MM coupling term, H_1, and the inter-MM coupling term, H_2. The coupling energy between the nearest-neighbor atoms in the same MM is denoted by t_1, and its contribution to the Hamiltonian is:

$$H_1 = -t_1 \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} \tag{S.4}$$

The interaction energy between the neighboring atoms from different MMs is denoted by t_2, and its contribution to the Hamiltonian is described by:

$$H_2 = -t_2 \begin{bmatrix} 0 & 0 & 0 & e^{i\vec{k}\cdot\vec{a}_1} & 0 & 0 \\ 0 & 0 & 0 & 0 & e^{i\vec{k}\cdot\vec{a}_2} & 0 \\ 0 & 0 & 0 & 0 & 0 & e^{i\vec{k}\cdot(\vec{a}_2-\vec{a}_1)} \\ e^{-i\vec{k}\cdot\vec{a}_1} & 0 & 0 & 0 & 0 & 0 \\ 0 & e^{-i\vec{k}\cdot\vec{a}_2} & 0 & 0 & 0 & 0 \\ 0 & 0 & e^{-i\vec{k}\cdot(\vec{a}_2-\vec{a}_1)} & 0 & 0 & 0 \end{bmatrix}, \tag{S.5}$$

where the $e^{i\vec{k}\vec{r}_{XY}}$ terms in the interaction between the X and Y atoms in neighboring MMs (the centers of the MMs are separated by the vector $\vec{r}_{XY}$) appear as a result of the Floquet periodic boundary conditions. The full interaction Hamiltonian $H_W = H_1 + H_2$ is given by:

$$H_W = -\begin{bmatrix} 0 & t_1 & 0 & t_2 e^{i\vec{k}\cdot\vec{a}_1} & 0 & t_1 \\ t_1 & 0 & t_1 & 0 & t_2 e^{i\vec{k}\cdot\vec{a}_2} & 0 \\ 0 & t_1 & 0 & t_1 & 0 & t_2 e^{i\vec{k}\cdot(\vec{a}_2-\vec{a}_1)} \\ t_2 e^{-i\vec{k}\cdot\vec{a}_1} & 0 & t_1 & 0 & t_1 & 0 \\ 0 & t_2 e^{-i\vec{k}\cdot\vec{a}_2} & 0 & t_1 & 0 & t_1 \\ t_1 & 0 & t_2 e^{-i\vec{k}\cdot(\vec{a}_2-\vec{a}_1)} & 0 & t_1 & 0 \end{bmatrix}. \tag{S.6}$$

Let us first analyze the system at the Γ point ($\vec{k} = \vec{0}$). The eigen-energies are computed to be: $E_s = -2t_1 - t_2$, $E_p = -t_1 + t_2$, $E_d = t_1 - t_2$, and $E_f = 2t_1 + t_2$. They correspond to the eigen-states of the system:

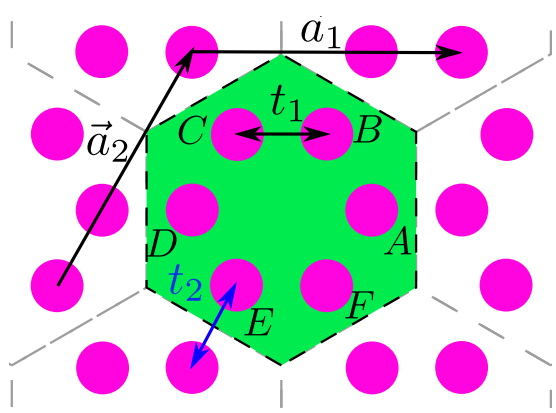

FIG. S2. Meta-molecules (MMs) made of six silicon pillars forming the triangular lattice of our system. A unit cell containing a single MM is shown in green. The lattice vectors are given by: $a_1 = [\sqrt{3}; 0]a$ and $a_2 = [\sqrt{3}/2; 3/2]a$. The intra-MM and inter-MM coupling coefficients are represented by t_1 and t_2, respectively.

$$|s\rangle = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad |p_+\rangle = \begin{bmatrix} 1 \\ e^{\frac{i\pi}{3}} \\ e^{\frac{2i\pi}{3}} \\ e^{\frac{3i\pi}{3}} \\ e^{\frac{4i\pi}{3}} \\ e^{\frac{5i\pi}{3}} \end{bmatrix}, \quad |d_+\rangle = \begin{bmatrix} 1 \\ e^{\frac{2i\pi}{3}} \\ e^{\frac{4i\pi}{3}} \\ e^{\frac{6i\pi}{3}} \\ e^{\frac{8i\pi}{3}} \\ e^{\frac{10i\pi}{3}} \end{bmatrix}, \quad |p_-\rangle = \begin{bmatrix} 1 \\ e^{\frac{-i\pi}{3}} \\ e^{\frac{-2i\pi}{3}} \\ e^{\frac{-3i\pi}{3}} \\ e^{\frac{-4i\pi}{3}} \\ e^{\frac{-5i\pi}{3}} \end{bmatrix}, \quad |d_-\rangle = \begin{bmatrix} 1 \\ e^{\frac{-2i\pi}{3}} \\ e^{\frac{-4i\pi}{3}} \\ e^{\frac{-6i\pi}{3}} \\ e^{\frac{-8i\pi}{3}} \\ e^{\frac{-10i\pi}{3}} \end{bmatrix}, \quad |f\rangle = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}. \tag{S.7}$$

The energy levels E_p and E_d are doubly degenerate. The eigen-states of the system at the Γ point with the lowest and highest energies resemble the atomic orbitals s and f, respectively. The eigen-states with intermediate energies, have a spin character and can be represented as a linear combination of atomic orbitals p_x, p_y, d_{xy}, and $d_{x^2-y^2}$: $p_\pm = \frac{1}{\sqrt{2}}(p_x \pm ip_y)$, $d_\pm = \frac{1}{\sqrt{2}}(d_{x^2+y^2} \pm id_{xy})$. The field distributions for the eigen-states $|p\pm\rangle$ have the phase $e^{\pm i\phi}$, where the angle ϕ is measured in the coordinate frame centered in the middle of the unit cell. These states have a strong p character. Similarly, field distributions for the eigen-states $|d\pm\rangle$ have the phase $e^{\pm 2i\phi}$, and these states have a strong d character. Both p and d states possess rotating (vortex-like) phase distributions analogous to the orbital angular momentum (pseudo-spin). The minus and plus sign corresponds to the clockwise (spin-up) and counterclockwise (spin-down) phase rotation direction, respectively.

In order to investigate the interaction between the spin-up and spin-down states, we express the Hamiltonian H_W in the basis of $|s\rangle$, $|p_\pm\rangle$, $|d_\pm\rangle$, and $|f\rangle$ orbitals. The new Hamiltonian is obtained by a standard procedure of change of the basis and is given by:

$$H_{\text{spdf}} = M^{-1} H_W M,$$

where M is a 6×6 matrix of the form:

$$M = \Big[\ |s\rangle;\ |p_+\rangle;\ |d_+\rangle;\ |p_-\rangle;\ |d_-\rangle;\ |f\rangle\ \Big],$$

whose columns represents the new basis vectors (s, $p_\pm$, $d_\pm$, f) in terms of the old basis vectors (Wannier functions). The Hamiltonian in the new basis has the form:

$$H_{\text{spdf}} = \begin{bmatrix} -2t_1 - t_2 + \frac{3}{4}a^2 t_2|\vec{k}|^2 & \frac{\sqrt{3}}{2}at_2(ik_x - k_y) & 0 & \frac{\sqrt{3}}{2}at_2(ik_x + k_y) & 0 & 0 \\ -\frac{\sqrt{3}}{2}at_2(ik_x + k_y) & t_2 - t_1 - \frac{3}{4}a^2 t_2|\vec{k}|^2 & -\frac{\sqrt{3}}{2}at_2(ik_x - k_y) & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{2}at_2(ik_x + k_y) & t_1 - t_2 + \frac{3}{4}a^2 t_2|\vec{k}|^2 & 0 & 0 & \frac{\sqrt{3}}{2}at_2(ik_x - k_y) \\ -\frac{\sqrt{3}}{2}at_2(ik_x - k_y) & 0 & 0 & t_2 - t_1 - \frac{3}{4}a^2 t_2|\vec{k}|^2 & -\frac{\sqrt{3}}{2}at_2(ik_x + k_y) & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}}{2}at_2(ik_x - k_y) & t_1 - t_2 + \frac{3}{4}a^2 t_2|\vec{k}|^2 & \frac{\sqrt{3}}{2}at_2(ik_x + k_y) \\ 0 & 0 & -\frac{\sqrt{3}}{2}at_2(ik_x + k_y) & 0 & -\frac{\sqrt{3}}{2}at_2(ik_x - k_y) & 2t_1 + t_2 - \frac{3}{4}a^2 t_2|\vec{k}|^2 \end{bmatrix} \tag{S.8}$$

where the terms $e^{i\vec{k}\vec{r}_{XY}}$ have been expressed using the Taylor expansion up to the terms linear (quadratic) in k_x and k_y in the off-diagonal (diagonal) elements.

Our main interest is to investigate the interaction between the spin states $|p_\pm\rangle$ and $|d_\pm\rangle$ because their interaction leads to the topological protection in the system. Therefore, in the following, we will consider a reduced 4×4 Hamiltonian containing only contributions from $|p_\pm\rangle$ and $d_\pm\rangle$ states [highlighted in Eq. (S.8)]. This Hamiltonian has a block diagonal form [5]:

$$H_{\text{pd}} = \begin{bmatrix} H_+ & 0 \\ 0 & H_- \end{bmatrix}, \tag{S.9}$$

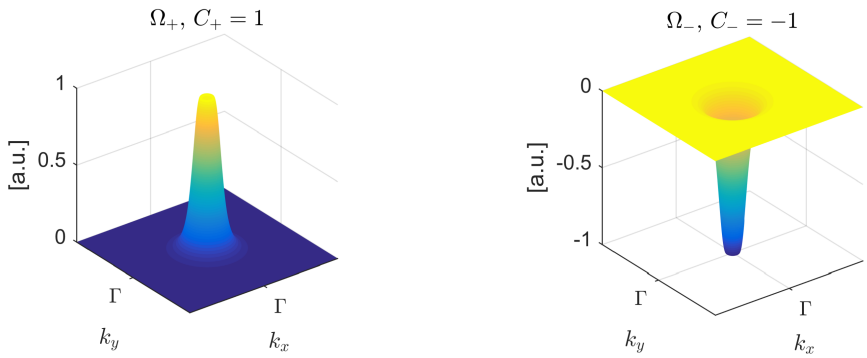

FIG. S3. Berry curvatures $\Omega_\pm$ and Chern numbers $C_\pm$ of corresponding bands in the vicinity of the Γ point for the system with $R = 0.359a_0$ and $n_{\mathrm{bg}} = 1.51$.

where $H_\pm = \frac{\sqrt{3}}{2}t_2 a(k_x\sigma_y \pm k_y\sigma_x) + (t_2 - t_1 - \frac{3}{4}a^2 t_2|\vec{k}|^2)\sigma_z$, and σ_x, σ_y, and σ_z are the Pauli matrices.

The Hamiltonian H_{pd} has the same form as the Bernevig-Hughes-Zhang Hamiltonian describing the quantum spin-Hall effect in a quantum well systems [6]. Thus, our structure is capable of supporting optical pseudo-spin states. The term $t_1 - t_2$ is proportional to the energy difference at the Γ point between the $|p_\pm\rangle$, $|d_\pm\rangle$ states. Depending on the sign of $t_1 - t_2$, the character of the bands might be inverted leading to a nontrivial topological properties of the system. For $t_1 > t_2$, the intra-MM coupling dominates and the energy of $|p_\pm\rangle$ bands is lower than the energy of $|d_\pm\rangle$ states, leading to the trivial state of the system. On the contrary, for $t_2 > t_1$, the inter-MM coupling is dominant, leading to inversion to the $|p_\pm\rangle$ and $|d_\pm\rangle$ bands at the Γ point and resulting in non-trivial topological properties [1]. When $t_2 = t_1$, the p and d bands are degenerate at the Γ point, resulting in the formation of the Dirac cone.

Topological properties of any structure can be inferred from a topological invariant called Chern number. Chern number describes the global [computed in the entire first Brillouin zone (BZ)] topological characteristics of the field distributions in a given energy band. In order to compute the Chern number, first we calculate the dispersion of light propagating in our topological structure. Using the PWE method, we obtain the $\omega(\vec{k})$ surfaces for the first 6 bands of the system together with the corresponding electric field distributions $E_z(\vec{r},\vec{k}) = \psi_k$. As mentioned above, we are interested only in states $|p_\pm\rangle$ and $|d_\pm\rangle$ showing the pseudo-spin character. For each of the energy bands, we compute the Berry connection $\vec{A}^\pm = [A_x^\pm; A_y^\pm] = \iint_{\text{unit cell}} \psi_k^{\pm *}\nabla_k\psi_k^\pm \,\mathrm{d}x\mathrm{d}y$, where minus and plus signs correspond to the spin-up and spin-down states, respectively, and ∇_k denotes a two-dimensional gradient operator in the k-space. Based on the Berry connection, we can compute the Berry curvature as:

$$\Omega_\pm = \frac{\partial A_y^\pm}{\partial k_x} - \frac{\partial A_x^\pm}{\partial k_y}, \tag{S.10}$$

Plots of the Berry curvature in the vicinity of the Γ point for our topological structure are shown in Fig. S3. Finally, the Chern number is calculated as:

$$C_\pm = \frac{1}{2\pi i}\iint_{1^{\mathrm{st}}\mathrm{BZ}} \Omega_\pm \mathrm{d}k_x\mathrm{d}k_y. \tag{S.11}$$

In our numerical approach, the eigen-spectrum of the Hamiltonian H_{pd} given by Eq. (S.9) is fitted to the $\omega(\vec{k})$ curves obtained from the PWE method allowing us to determine the t_1 and t_2 parameters. Then, the eigen-states of H_{pd} are used to compute the Berry curvature and the Chern number using the approach presented in Ref. [7].

Reconfigurability enabled by liquid crystals

Liquid crystals (LCs) are a state of matter that simultaneously exhibit properties of liquids (fluidity) and crystals (anisotropy resulting from orientation order of the molecules). LCs offer a possibility of refractive index tuning with unprecedented amplitude reaching 10% [8]. The elongated nematic LC molecules, schematically shown in Fig. S4(a), are preferentially aligned along a particular direction, called director, resulting in uniaxial anisotropy of the LC medium. The refractive index experienced by the extraordinarily polarized eigen-wave, $n_e(\theta)$, changes depending on

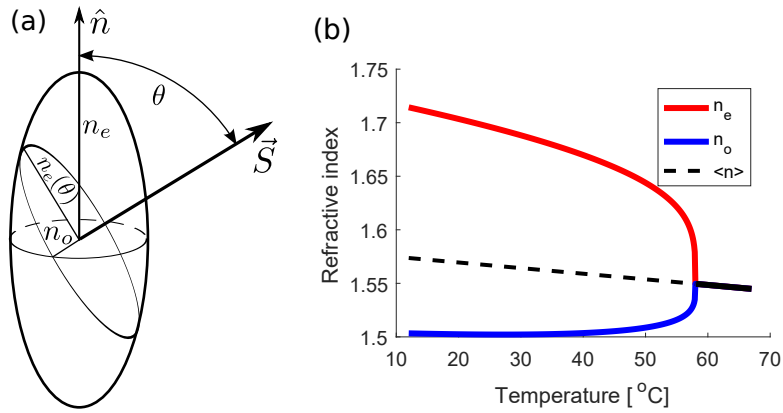

FIG. S4. (a) Schematic illustration of a liquid crystal molecule showing the refractive indices along the long (director) axis n_e, along the short axis (ordinary index) n_o, and the extraordinary refractive index $n_e(\theta)$. The angle between the director axis $\hat{n}$ and the light propagation direction $\vec{S}$ is called θ. (b) Dependence of the refractive indices n_o and n_e on temperature T for the $E7$ liquid crystal at 1550 nm obtained using model from Ref. [9].

the angle θ between the director $\hat{n}$ and the light propagation direction $\vec{S}$:

$$n_e(\theta) = \frac{n_e n_o}{\left[n_e^2 \cos^2(\theta) + n_o^2 \sin^2(\theta)\right]^{1/2}}, \tag{S.12}$$

where n_o denotes the ordinary refractive index and n_e is the index of the wave polarized along the optical axis of the molecule. In this paper, we consider only the two extreme orientations of the LC molecules: (i) perpendicular to the PC pillars, for which the TM-polarized wave experiences the background index $n_{\rm bg} = n_o$; and (ii) parallel to the PC pillars, for which $\theta = 90^\circ$ and $n_{\rm bg} = n_e$, as explained below.

The PC structure studied in the main text of the manuscript can be bound from top and bottom by metal plates, as illustrated in Fig. 1(a) in the main text of the manuscript, that serve two goals. Firstly, for sufficiently small separation between the plates, the field distribution inside the PC structure remains uniform in the z-direction. Alternatively, the electrodes can be located at a distance from the PC slab. The space between electrodes and the slab can be filled with the LC which has much lower index than the PC itself. In this case, the light confinement stems from the total internal reflection between the high-index silicon slab and the low-index cladding. In both cases, these 3-dimensional (3D) systems can be well approximated using 2D z-invariant modeling. This approximated treatment does not take into account the losses resulting from light penetrating into the metal regions and additional optimization is required to minimize these losses. Our 2D PC supports transverse-electric and transverse-magnetic modes. Here, we are only interested in the TM modes because the studied structure exhibits topological properties only for the this light polarization [1]. The TM mode has non-zero magnetic field components H_x and H_y lying in the x–y plane perpendicular to the silicon pillars and an electric field component E_z along the z-axis parallel to the pillars.

Secondly, the conductive plates serve as electrodes supplying an external electric field along the z-direction that allows for LC reorientation [8, 10]. Due to the birefringent nature of LCs, the electric field exerts a torque on the LC molecules and, in case of positive birefringence ($\Delta n = n_e - n_o > 0$), reorients them along the field direction. In our structure, the surface of the electrodes is prepared in such a way that the anchoring energy holds the surface molecules aligned parallel to the electrodes. In the absence of the electric field, the bulk molecules align along the surface molecules to minimize the global LC free energy, as shown in Fig. 1(d) in the main text of the manuscript. In this case, the TM-polarized wave is the ordinary wave and it experiences the background refractive index $n_{\rm bg} = n_o$, irrespectively of the propagation direction in the x–y plane and the in-plane director orientation. Application of an external electric field, with the amplitude above the threshold of the Fréedericksz transition, results in rotation of the LC molecules [11], as shown in Fig. 1(b) in the main text of the manuscript. Then, the director lies along the z-axis and is perpendicular to the light propagation direction ($\theta = 90^\circ$). In this configuration, the background refractive index experienced by the TM-polarized wave (extraordinary wave) is $n_{\rm bg} = n_e$, irrespectively of the in-plane propagation direction.

Orientations of the LC molecules at intermediate values of the angle θ, in addition to being difficult to achieve experimentally, also break the reflection symmetry of the PC structure with respect to the x–y plane. As a result, the

TM-polarized wave excites both the ordinary and the extraordinary waves, and experiences polarization rotation due to the birefringence of the LC background. Systems with intermediate angles θ are not considered here and require further investigation.

Rotation of the LC molecules takes a finite amount of time due to their inertia and viscosity [12]. Typically, this switching time is of the order of microseconds allowing for device operation at MHz frequencies. Recently, sub-microsecond switching times were also reported [13–15]. In this paper, we use the $E7$ LC at the room temperature T = 25°C which permits background index changes in the range between n_o = 1.51 and n_e = 1.69 [9, 10]. Additional tunability can be added exploiting the fact that the values of n_o and n_e can be modified by temperature tuning, as seen in Fig. S4(b). For low temperatures, the LC birefringence is high, and the reorientation of the molecules leads to large changes in the background refractive index. At higher temperatures, the tunability is possible only in a limited range of background refractive indices, and therefore only a part of the parameter space shown in Figs. 4(b)–(d) in the main text of the manuscript can be explored. Finally, for the temperatures above the nematic/isotropic phase transition point, no tunability is possible, as the LC becomes an isotropic liquid.

[1] Wu, L. H. & Hu, X. Scheme for achieving a topological photonic crystal by using dielectric material. *Phys. Rev. Lett.* **114**, 223901 (2015).

[2] Barik, S., Miyake, H., DeGottardi, W., Waks, E. & Hafezi, M. Two-dimensionally confined topological edge states in photonic crystals. *New J. Phys.* **18**, 113013 (2016).

[3] Hatsugai, Y. Chern number and edge states in the integer quantum Hall-effect. *Phys. Rev. Lett.* **71**, 3697–3700 (1993).

[4] Xu, L., Wang, H. X., Xu, Y. D., Chen, H. Y. & Jiang, J. H. Accidental degeneracy in photonic bands and topological phase transitions in two-dimensional core-shell dielectric photonic crystals. *Opt. Express* **24**, 18059–18071 (2016).

[5] Wu, L.-H. & Hu, X. Topological properties of electrons in honeycomb lattice with detuned hopping energy. *Sci. Rep.* **6**, 24347 (2016).

[6] Bernevig, B. A., Hughes, T. L. & Zhang, S.-C. Quantum spin Hall effect and topological phase transition in HgTe quantum wells. *Science* **314**, 1757–1761 (2006).

[7] Fukui, T., Hatsugai, Y. & Suzuki, H. Chern numbers in discretized Brillouin zone: Efficient method of computing (spin) Hall conductances. *J. Phys. Soc. Jpn.* **74**, 1674–1677 (2005).

[8] Khoo, I.-C. & Wu, S.-T. *Optics and nonlinear optics of liquid crystals.* Series in nonlinear optics (World Scientific, 1993).

[9] Li, J., Wu, S. T., Brugioni, S., Meucci, R. & Faetti, S. Infrared refractive indices of liquid crystals. *J. Appl. Phys.* **97**, 073501 (2005).

[10] Weirich, J. *et al.* Liquid crystal parameter analysis for tunable photonic bandgap fiber devices. *Opt. Express* **18**, 4074–4087 (2010).

[11] Frisken, B. J. & Palffy-Muhoray, P. Freedericksz transitions in nematic liquid crystals: The effects of an in-plane electric field. *Phys. Rev. A* **40**, 6099–6102 (1989).

[12] Silva, A. T., Santos, M. A. F. D., Lenzi, E. K., Evangelista, L. R. & Zola, R. S. Modeling nematic liquid crystals: Analytical solution for the balance of torques equation with moment of inertia and surface viscosity. *Mol. Cryst. Liq. Cryst.* **576**, 32–41 (2013).

[13] Khoo, I. C., Liou, J., Stinger, M. V. & Zhao, S. Ultrafast all-optical switching with transparent and absorptive nematic liquid crystals–Implications in tunable metamaterials. *Mol. Cryst. Liq. Cryst.* **543**, 151/[917]–159/[925] (2011).

[14] Borshch, V., Shiyanovskii, S. V. & Lavrentovich, O. D. Nanosecond electro-optic switching of a liquid crystal. *Phys. Rev. Lett.* **111**, 107802 (2013).

[15] Mohammadimasoudi, M., Beeckman, J., Shin, J., Lee, K. & Neyts, K. Widely tunable chiral nematic liquid crystal optical filter with microsecond switching time. *Opt. Express* **22**, 19098–19107 (2014).